Programming Fundamentals Lab 05

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Lab 05 Assignment
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Task 5: Nested Switch-Case for Country Listing

Write a C program that implements the Nested Switch-Case for a School Grading System as described.

Instructions:

1. Prompt the user to enter a course number (1, 2, or 3).
2. Based on the course number, prompt for a letter grade (‘A’, ‘B’, ‘C’, ‘D’, or ‘F’).
3. Using a nested switch-case structure, print the appropriate message for the student’s performance in that specific course.
4. Include a default case in both the outer and inner switch statements to handle invalid inputs.

C Code:

```
#include <stdio.h>
int main()
{
   char num, grade;
   printf("Enter your course number(1, 2 or 3).\n");
   scanf(" %c", &num);
   switch (num)
   {
   case '1':
      printf("Enter your grade (A, B, C, D, or F).\n");
      scanf(" %c", &grade);
      switch (grade)
      {
```

```
    case 'A':
        printf("Your performance was Excellent.\n");
        break;
    case 'B':
        printf("Your performance was Good.\n");
        break;
    case 'C':
        printf("Your performance was mediocre.\n");
        break;
    case 'D':
        printf("Your performance was Lacking, More practice and study needed.\n");
        break;
    case 'F':
        printf("You Failed, The course needs to be repeated.\n");
        break;
    default:
        printf("invalid grade input, please try again.\n");
        break;
    }
    break;
case '2':
    printf("Enter your grade (A, B, C, D, or F).\n");
    scanf(" %c", &grade);
    switch (grade)
    {
    case 'A':
```

```
        printf("Your performance was Excellent.\n");
        break;
    case 'B':
        printf("Your performance was Good.\n");
        break;
    case 'C':
        printf("Your performance was mediocre.\n");
        break;
    case 'D':
        printf("Your performance was Lacking, More practice and study needed.\n");
        break;
    case 'F':
        printf("You Failed, The course needs to be repeated.\n");
        break;
    default:
         printf("invalid grade input, please try again.\n");
        break;
    }
    break;
case '3':
    printf("Enter your grade (A, B, C, D, or F).\n");
    scanf(" %c", &grade);
    switch (grade)
    {
    case 'A':
        printf("Your performance was Excellent.\n");
```

```
            break;
        case 'B':
            printf("Your performance was Good.\n");
            break;
        case 'C':
            printf("Your performance was mediocre.\n");
            break;
        case 'D':
            printf("Your performance was Lacking, More practice and study needed.\n");
            break;
        case 'F':
            printf("You Failed, The course needs to be repeated.\n");
            break;
        default:
            printf("invalid grade input, please try again.\n");
            break;
        }
        break;
    default:
        printf("Invalid number input, please try again.\n");
        break;
    }
return 0;
}
```

```
Enter your course number(1, 2 or 3).
1
Enter your grade (A, B, C, D, or F).
A
Your performance was Excellent.
```

```
Enter your course number(1, 2 or 3).
2
Enter your grade (A, B, C, D, or F).
B
Your performance was Good.
```

```
Enter your course number(1, 2 or 3).
3
Enter your grade (A, B, C, D, or F).
F
You Failed, The course needs to be repeated.
```

Task 6: Nested Ternary Operator for Maximum Value

Use nested ternary operators to find the maximum of three numbers.

Instructions:

1. Take three integer inputs from the user (x, y, z).

2. Use nested ternary operators to find and display the largest of the three numbers.

C Code:

```
#include <stdio.h>

int main()

{

   int num1, num2, num3;

   printf("Enter the first number.\n");

   scanf("%d", &num1);

   printf("Enter the second number.\n");

   scanf("%d", &num2);

   printf("Enter the third number.\n");

   scanf("%d", &num3);

   (num1>num2)?(num1>num3)?printf("%d is the largest.", num1):printf("%d is the largest.", num3):(num2>num3)?printf("%d is the largest.", num2):printf("%d is the largest.", num3);

return 0;

}
```

```
Enter the first number.
9
Enter the second number.
3
Enter the third number.
6
9 is the largest.
```

```
Enter the first number.
3
Enter the second number.
8
Enter the third number.
1
8 is the largest.
```

```
Enter the first number.
2
Enter the second number.
5
Enter the third number.
7
7 is the largest.
```

Task 7: Arithmetic Operators

Write a program that accepts two integers and performs various arithmetic operations on them.

Instructions:

1. Take two integer inputs from the user (a, b).

2. Use arithmetic operators to calculate and display:

o Sum of a and b

o Difference between a and b

o Product of a and b

o Quotient of a and b

o Remainder when a is divided by b

C Code:

```
#include <stdio.h>

int main()

{

   int a, b;

   printf("Enter the first number.\n");

   scanf("%d", &a);

   printf("Enter the second number.\n");

   scanf("%d", &b);

   printf("The sum of %d and %d is %d.\n", a, b, a+b);

   printf("The difference of %d and %d is %d.\n", a, b, a-b);

   printf("The product of %d and %d is %d.\n", a, b, a*b);

   printf("The quotient of %d and %d is %d.\n", a, b, a/b);

   printf("The remainder when %d divided by %d is %d.\n", a, b, a%b);
```

```
return 0;

}
```

```
Enter the first number.
6
Enter the second number.
4
The sum of 6 and 4 is 10.
The difference of 6 and 4 is 2.
The product of 6 and 4 is 24.
The quotient of 6 and 4 is 1.
The remainder when 6 divided by 4 is 2.
```

Task 8: Relational Operators

Demonstrate the use of relational operators by comparing two user-entered integers.

Instructions:

1. Accept two integer inputs from the user (a, b).

2. Use relational operators to compare the two values and display the results of the following comparisons:

o a == b

o a > b

o a < b

o a != b

o a >= b

o a <= b

Example Input:

a = 10, b = 5

Expected Output:

a > b: 1 (True)

a < b: 0 (False)

a == b: 0 (False)

C Code:

```
#include <stdio.h>

int main()

{

   int A, B;

   printf("Enter the First number.\n");
```

```
    scanf("%d",&A);
    printf("Enter the Second number.\n");
    scanf("%d",&B);
    printf("The result of using the realational operator == is %d\n", A==B);
    printf("The result of using the realational operator > is %d\n", A>B);
    printf("The result of using the realational operator < is %d\n", A<B);
    printf("The result of using the realational operator != is %d\n", A!=B);
    printf("The result of using the realational operator >= is %d\n", A>=B);
    printf("The result of using the realational operator <= is %d\n", A<=B);
return 0;
}
```

```
Enter the First number.
4
Enter the Second number.
8
The result of using the realational operator == is 0
The result of using the realational operator > is 0
The result of using the realational operator < is 1
The result of using the realational operator != is 1
The result of using the realational operator >= is 0
The result of using the realational operator <= is 1
```

Task 9: Use Bitwise Operators

Practice using bitwise operators by writing a program that performs bitwise operations on two integers.

Instructions:

1. Take two integers (a and b) as input from the user.

2. Perform and display the following bitwise operations:

o a & b (Bitwise AND)

o a | b (Bitwise OR)

o a ^ b (Bitwise XOR)

o ~a (Bitwise NOT)

o a << 1 (Shift left)

o a >> 1 (Shift right)

C Code:

```
#include <stdio.h>

int main()

{

   int A, B;

   printf("Enter the First number.\n");

   scanf("%d",&A);

   printf("Enter the Second number.\n");

   scanf("%d",&B);

   printf("The result of using the BitWise operator & is %d\n", A&B);

   printf("The result of using the BitWise operator | is %d\n", A|B);

   printf("The result of using the BitWise operator ~ on %d is %d\n", A, ~A);
```

```
    printf("The result of using the BitWise operator << is %d\n", A<<1);
    printf("The result of using the BitWise operator >> is %d\n", A>>1);
return 0;
}
```

```
Enter the First number.
5
Enter the Second number.
8
The result of using the BitWise operator & is 0
The result of using the BitWise operator | is 13
The result of using the BitWise operator ~ on 5 is -6
The result of using the BitWise operator << is 10
The result of using the BitWise operator >> is 2
```

Task 10: Nested If-Else for Largest of Three Numbers

Practice using nested if-else statements by writing a program that finds the largest of three user-

entered numbers.

Instructions:

1. Take three integer inputs from the user (x, y, z).

2. Use a nested if-else structure to determine which number is the largest.

3. Display the largest number.

C Code:

```
#include <stdio.h>

int main()

{

   int x, y, z;

   printf("Enter the First number.\n");

   scanf("%d", &x);

   printf("Enter the Second number.\n");

   scanf("%d", &y);

   printf("Enter the Third number.\n");

   scanf("%d", &z);

   if (x>y && x>z)

   {

      printf("%d is the largest.\n", x);

   }

   else if (y>x && y>z)
```

```
    {
        printf("%d is the largest.\n", y);
    }
    else if (z>x && z>y)
    {
        printf("%d is the largest.\n", z);
    }
    else
    {
        printf("The numbers are equal.\n");
    }
return 0;
}
```

```
Enter the First number.
7
Enter the Second number.
1
Enter the Third number.
3
7 is the largest.
```

```
Enter the First number.
2
Enter the Second number.
9
Enter the Third number.
5
9 is the largest.
```

```
Enter the First number.
3
Enter the Second number.
5
Enter the Third number.
8
8 is the largest.
```